



Spring MVC with Boot

- ♦ You can develop Spring MVC application with Spring Boot in two ways.
 - Develop as **FAT JAR** file.
 - Develop as **WAR** file
- ♦ FAT Jar is self executable application which contains **Application Code + Embbeded Container**
- ♦ You can execute the FAT jar as follows.
 - Java -jar HelloApp.jar
- ♦ WAR file is deployable application which can be deployed in any External Container to run.

demo1-mvc-boot: Spring MVC with Boot (using JSP)

Working Steps:

A) Setup the Project

1) Create the maven Project.

- a) Open the Eclipse required Workspace
- b) Select New -> Maven Project.
- c) Check the checkbox for create Simple Project(skip the Archetype selection) and Click on Next Button.
- d) Provide the following
 - Group Id: com.coursecube.springboot
 - Artifact Id : **demo1-mvc-boot**
 - Version : 1.0
 - packaging : jar
 - Name : **demo1-mvc-boot**and Click On finish button.

2) Remove the following two Source Folders.

- a. src/test/java
- b. src/test/resources

3) Create application.properties under src/main/resources

4) Update application.properties as follows

server.port=12345

5) Create the Source Folder called **src/main/webapp**

6) Create the Folder called **WEB-INF** under src/main/webapp



7) Create the Folder called **myjsps** under WEB-INF

8) Update pom.xml by adding the following dependencies.

```
<parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>2.1.16.RELEASE</version>
</parent>

<dependencies>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-web</artifactId>
    </dependency>

    <dependency>
        <groupId>org.apache.tomcat.embed</groupId>
        <artifactId>tomcat-embed-jasper</artifactId>
    </dependency>

    <dependency>
        <groupId>jstl</groupId>
        <artifactId>jstl</artifactId>
        <version>1.2</version>
    </dependency>

    <dependency>
        <groupId>>taglibs</groupId>
        <artifactId>standard</artifactId>
        <version>1.1.2</version>
    </dependency>

    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-devtools</artifactId>
    </dependency>
</dependencies>
```

9) You can see the Error in Maven Project.

Fix the Error by updating the Maven Project.

Select the Project , right click and then select Maven -> Update Project.



- 10) Create the Package called **com.jlcindia.springboot** under src/main/java
- 11) Write Bean Configuration class – **JLCWebConfig.java**
- 12) Configure View Resolver in **JLCWebConfig.java**

@SpringBootApplication

```
public class JLCWebConfig implements WebMvcConfigurer{
```

@Bean

```
    public ViewResolver viewResolver() {  
        InternalResourceViewResolver resolver = new InternalResourceViewResolver();  
        resolver.setPrefix("/WEB-INF/myjsps/");  
        resolver.setSuffix(".jsp");  
        resolver.setViewClass(JstlView.class);  
        return resolver;  
    }  
}
```

- 13) Write Boot Main Class – MyBootApplication.java

@SpringBootApplication

```
public class MyBootApplication {  
    public static void main(String[] args) {  
        SpringApplication.run(StartMyBootApplication.class, args);  
    }  
}
```

- 14) Run the Boot Application

You will see – Whitelabel Error Page means DS is Ready.

B)Develop the Usecase : show index page

http://localhost:12345/ => You are sending the request for /

- 15) Write index.jsp under WEB-INF/myjsps folder.

```
<!DOCTYPE html>  
<html>  
<body>  
<h1> Welcome to Java Learning Center !!!</h1>  
</body>  
</html>
```



16) Write IndexController.java

```
@Controller
public class IndexController {
    @GetMapping("/")
    public String showIndexPage() {
        System.out.println("IndexController - showIndexPage()");
        return "index";
    }
}
```

17) Run the Boot Application

18) Open the browser and hit <http://localhost:12345/>
If every thing fine you will see index.jsp

C)Enable BootStarp:

19) Specify the Dependencies for JQuery and Bootstarp

```
<dependency>
    <groupId>org.webjars</groupId>
    <artifactId>bootstrap</artifactId>
    <version>4.3.1</version>
</dependency>

<dependency>
    <groupId>org.webjars</groupId>
    <artifactId>jquery</artifactId>
    <version>3.4.1</version>
</dependency>
```

20) Observe the following

a) bootstrap-4.3.1.jar

META-INF=> resources => webjars => bootstrap => 4.3.1 => css => bootstrap.min.css

META-INF=> resources => webjars => bootstrap => 4.3.1 => js => bootstrap.min.js

b) jquery-3.4.1.jar =>

META-INF=> resources => webjars => jquery => 3.4.1 => jquery.min.js



- 21) Add the resource handler bean (to make your resources available in class path) in JLCWebConfig

```
@Override
public void addResourceHandlers(ResourceHandlerRegistry registry) {
    registry.addResourceHandler("/webjars/**").addResourceLocations("classpath:/META-INF/resources/webjars/");
}
```

- 22) Include the css and js files in index.jsp.

```
<link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">

<script src="webjars/jquery/3.4.1/jquery.min.js"></script>
<script src="webjars/bootstrap/4.3.1/js/bootstrap.min.js"></script>
```

- 23) Use Some Bootstrap Classes in index.jsp

```
<h1 class="text-center"> Welcome to Java Learning Center !!!</h1>
```

- 24) Run the Boot Application and Test

- 25) Open the browser and hit <http://localhost:12345/>

If every thing fine you will see Bootstrap Css styles applied in index.jsp

D)Develop the Usecase : show books

<http://localhost:12345/showBooks>

- 26) Add hyperlink in index.jsp

```
<h2 class="text-center"> <a href="showBooks"> Show Books </a></h2>
```

- 27) Write Book.java under **com.jlcindia.springboot**

- 28) Write BooksController.java under **com.jlcindia.springboot**

- 29) Write booksList.jsp under **WEB-INF/myjsps**

- 30) Run the Boot Application and Test

Open the browser and hit <http://localhost:12345/>

demo1-mvc-boot: Files Required

1. index.jsp	2. booksList.jsp
3. IndexController.java	4. BooksController.java
5. Book.java	6. JLCWebConfig.java
7. MyBootApplication.java	8. application.properties
9. pom.xml	

Welcome to CourseCube !!!

Latest Books from Srinivas Dande

Book Id	Book Name	Author	Price	Publications		
101	Master Spring Boot	Srinivas Dande	10000.0	JLC	View Book	Edit Book
102	Master MicroServices	Srinivas Dande	10000.0	JLC	View Book	Edit Book
103	Master Angular	Srinivas Dande	10000.0	JLC	View Book	Edit Book
104	Master React	Srinivas Dande	10000.0	JLC	View Book	Edit Book
105	Master Java Full Stack	Srinivas Dande	10000.0	JLC	View Book	Edit Book

1) index.java

```

<!DOCTYPE html>
<html>
<head>
<title> JLCBookStore</title>
<link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<h1 class="text-center"> Welcome to Java Learning Center !!!</h1>
<br/>
<h2 class="text-center"> <a href="showBooks"> Show Books </a></h2>

<script src="webjars/jquery/3.4.1/jquery.min.js"></script>
<script src="webjars/bootstrap/4.3.1/js/bootstrap.min.js"></script>

</body>
</html>

```



2) BooksList.jsp

```
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>

<!DOCTYPE html>
<html>  <head>
<title> JLCBookStore</title>
<link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<div class="container">
<h1 class="text-center"> Welcome to Java Learning Center !!!</h1>

<div class="alert alert-danger" role="alert">
<h2 class="text-center"> Latest Books from Srinivas Dande </h2>
</div>

<br/>
<table class="table table-striped table-bordered table-hover" style="font-size:20px;">
<thead class="thead-light">
<tr>  <th> Book Id </th>
<th> Book Name </th>
<th> Author</th>
<th> Price </th>
<th> Publications</th>
<th> </th>
<th> </th>  </tr>
</thead>
<tbody>
<c:forEach var="mybook" items="${MyBooks}">
<tr>
<td> <a href="#"> ${mybook.bid} </a> </td>
<td> ${mybook.bname} </td>
<td> ${mybook.author} </td>
<td> ${mybook.price} </td>
<td> ${mybook.pub}</td>
<td> <button type="button" class="btn btn-success btn-lg">View Book</button> </td>
<td> <button type="button" class="btn btn-primary btn-lg">Edit Book</button> </td>
</tr>
</c:forEach>
</tbody>
</table>
</div>
<script src="webjars/jquery/3.4.1/jquery.min.js"></script>
<script src="webjars/bootstrap/4.3.1/js/bootstrap.min.js"></script>
</body>
</html>
```



3) IndexController.java

```
package com.jlcindia.springboot;

import org.springframework.stereotype.Controller;
import org.springframework.web.bind.annotation.GetMapping;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@Controller
public class IndexController {

    @GetMapping(value = "/")
    public String showIndexPage() {
        System.out.println("IndexController - showIndexPage()");
        return "index";
    }
}
```

4) BooksController.java

```
package com.jlcindia.springboot;

import java.util.ArrayList;
import java.util.List;
import javax.servlet.http.HttpSession;
import org.springframework.stereotype.Controller;
import org.springframework.web.bind.annotation.GetMapping;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@Controller
public class BookController {

    @GetMapping("/showBooks")
    public String getBooks(HttpSession session) {
        System.out.println(" BooksController - getBooks()");

        List<Book> blist=new ArrayList<>();

        Book mybook1=new Book(101,"Master Spring Boot","Srinivas Dande ",10000.0,"JLC");
        Book mybook2=new Book(102,"Master MicroServices","Srinivas Dande",10000.0,"JLC");
        Book mybook3=new Book(103,"Master Angular","Srinivas Dande",10000.0,"JLC");
        Book mybook4=new Book(104,"Master React ","Srinivas Dande",10000.0,"JLC");
        Book mybook5=new Book(105,"Master Java Full Stack","Srinivas Dande",10000.0,"JLC");
    }
}
```



```
blist.add(mybook1);
blist.add(mybook2);
blist.add(mybook3);
blist.add(mybook4);
blist.add(mybook5);

session.setAttribute("MyBooks", blist);

return "booksList";
}
}
```

5) Book.java

```
package com.jlcindia.springboot;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

public class Book {

    private int bid;
    private String bname;
    private String author;
    private double price;
    private String pub;

    public Book() {}
    public Book(int bid, String bname, String author, double price, String pub) {
        super();
        this.bid = bid;
        this.bname = bname;
        this.author = author;
        this.price = price;
        this.pub = pub;
    }
    //Setters and Getters
    //toString()
}
```



6) JLCWebConfig.java

```
package com.jlcindia.springboot;

import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.annotation.Bean;
import org.springframework.web.servlet.ViewResolver;
import org.springframework.web.servlet.config.annotation.ResourceHandlerRegistry;
import org.springframework.web.servlet.config.annotation.WebMvcConfigurer;
import org.springframework.web.servlet.view.InternalResourceViewResolver;
import org.springframework.web.servlet.view.JstlView;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@SpringBootApplication
public class JLCWebConfig implements WebMvcConfigurer{
    @Bean
    public ViewResolver viewResolver() {
        InternalResourceViewResolver resolver = new InternalResourceViewResolver();
        resolver.setPrefix("/WEB-INF/myjsps/");
        resolver.setSuffix(".jsp");
        resolver.setViewClass(JstlView.class);
        return resolver;
    }
    @Override
    public void addResourceHandlers(ResourceHandlerRegistry registry) {
        registry.addResourceHandler("/webjars/**")
        .addResourceLocations("classpath:/META-INF/resources/webjars/");
    }
}
```

7) MyBootApplication.java

```
package com.jlcindia.springboot;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@SpringBootApplication
public class MyBootApplication {
    public static void main(String as[]) {
        SpringApplication.run(MyBootApplication.class, as);
    }
}
```



8)application.properties

```
server.port=12345
```

9)pom.xml

```
<project ...>

<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.springboot</groupId>
<artifactId>demo1-mvc-boot</artifactId>
<version>1.0</version>
<name>demo1-mvc-boot</name>
<description>This is My Spring MVC App with Boot</description>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.5.3</version>
</parent>

<dependencies>
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>

<dependency>
<groupId>org.apache.tomcat.embed</groupId>
<artifactId>tomcat-embed-jasper</artifactId>
</dependency>

<dependency>
<groupId>jstl</groupId>
<artifactId>jstl</artifactId>
<version>1.2</version>
</dependency>

<dependency>
<groupId>taglibs</groupId>
<artifactId>standard</artifactId>
<version>1.1.2</version>
</dependency>
```



```
<dependency>
<groupId>org.webjars</groupId>
<artifactId>bootstrap</artifactId>
<version>4.3.1</version>
</dependency>

<dependency>
<groupId>org.webjars</groupId>
<artifactId>jquery</artifactId>
<version>3.4.1</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>

<build>
<finalName>myjlcapp</finalName>
<plugins>

<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>
</project>
```



demo2-mvc-boot: Spring MVC with Boot (using ThymeLeaf)

Working Steps:

- 1) Copy **demo1-mvc-boot** as **demo2-mvc-boot**
- 2) Remove the WEB-INF under src/main/webapp (So that all the JSP's will be deleted)
- 3) Remove **ViewResolver** Bean from JLCWebConfig.java
- 4) Create templates folder under src/main/resources
- 5) Place index.html and booksList.html under templates
 - a) index.html (new in Demo2)
 - b) booksList.html (new in Demo2)
- 6) Add the following property in application.properties

```
server.port=54321
spring.thymeleaf.cache=false
```
- 7) Remove the following dependencies from pom.xml

```
<dependency>
  <groupId>org.apache.tomcat.embed</groupId>
  <artifactId>tomcat-embed-jasper</artifactId>
</dependency>

<dependency>
  <groupId>jstl</groupId>
  <artifactId>jstl</artifactId>
  <version>1.2</version>
</dependency>

<dependency>
  <groupId>>taglibs</groupId>
  <artifactId>standard</artifactId>
  <version>1.1.2</version>
</dependency>
```
- 8) Add the dependency for ThymeLeaf in pom.xml

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-thymeleaf</artifactId>
</dependency>
```

- 9) Make sure that all the Files as given below are updated correctly.
- 10) Run the Boot Application
- 11) Open the browser and hit <http://localhost:54321/>

demo2-mvc-boot: Files Required

1. index.html	New in Demo2
2. booksList.html	New in Demo2
3. Book.java	Same as Demo1
4. IndexController.java	Same as Demo1
5. BooksController.java	Same as Demo1
6. JLCWebConfig.java	Updated in Demo2
7. StartMyBootApplication.java	Same as Demo1
8. application.properties	Updated in Demo2
9. pom.xml	Updated in Demo2

Welcome to CourseCube !!!

Latest Books from Srinivas Dande

Book Id	Book Name	Author	Price	Publications		
101	Master Spring Boot	Srinivas Dande	10000.0	JLC	View Book	Edit Book
102	Master MicroServices	Srinivas Dande	10000.0	JLC	View Book	Edit Book
103	Master Angular	Srinivas Dande	10000.0	JLC	View Book	Edit Book
104	Master React	Srinivas Dande	10000.0	JLC	View Book	Edit Book
105	Master Java Full Stack	Srinivas Dande	10000.0	JLC	View Book	Edit Book



1) index.html

```
<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
<title> JLCBookStore</title>
<link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>

<body>
<h1 class="text-center"> Welcome to JLCINDIA !!! </h1>
<br/>
<h2 class="text-center"> <a th:href="@{/showBooks}">Show Books</a></h2>

<script src="webjars/jquery/3.4.1/jquery.min.js"></script>
<script src="webjars/bootstrap/4.3.1/js/bootstrap.min.js"></script>

</body>
</html>
```

2) BooksList.html

```
<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
<title>JLC Bookstore</title>
<link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<div class="container">
<h1 class="text-center"> Welcome to JLCINDIA !!! </h1>

<div class="alert alert-primary" role="alert">
<h2 class="text-center"> Latest Books from Srinivas Dande</h2>
</div>

<br />
<table class="table table-striped table-bordered table-hover"
style="font-size: 20px;">
<thead class="thead-light">
<tr>
<th>Book Id</th>
<th>Book Name</th>
<th>Author</th>
<th>Price</th>
<th>Publications</th>
<th></th>
<th></th>
</thead>
```



```
</tr>
</thead>
<tbody>
<tr th:each="mybook : ${session.MyBooks}">
<td th:text="${mybook.bid}">      </td>
<td th:text="${mybook.bname}"></td>
<td th:text="${mybook.author}"></td>
<td th:text="${mybook.price}"></td>
<td th:text="${mybook.pub}"></td>
<td>
<button type="button" class="btn btn-success btn-lg">View
Book</button>
</td>
<td>
<button type="button" class="btn btn-danger btn-lg">Edit
Book</button>
</td>
</tr>
</tbody>
</table>
</div>

<script src="webjars/jquery/3.4.1/jquery.min.js"></script>
<script src="webjars/bootstrap/4.3.1/js/bootstrap.min.js"></script>

</body>
</html>
```

6) JLCWebConfig.java

```
package com.jlcindia.springboot;

import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.web.servlet.config.annotation.ResourceHandlerRegistry;
import org.springframework.web.servlet.config.annotation.WebMvcConfigurer;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@SpringBootApplication
public class JLCWebConfig implements WebMvcConfigurer{

    @Override
    public void addResourceHandlers(ResourceHandlerRegistry registry) {
        registry.addResourceHandler("/webjars/**")
        .addResourceLocations("classpath:/META-INF/resources/webjars/");
    }
}
```



8)application.properties

server.port=54321

spring.thymeleaf.cache=false

9)pom.xml

<project ...>

<modelVersion>4.0.0</modelVersion>

<groupId>**com.jlcindia.springboot**</groupId>

<artifactId>**demo2-mvc-boot**</artifactId>

<version>1.0</version>

<name>**demo2-mvc-boot**</name>

<description>This is My Spring MVC App with Boot</description>

<parent>

<groupId>org.springframework.boot</groupId>

<artifactId>spring-boot-starter-parent</artifactId>

<version>**2.5.3**</version>

</parent>

<dependencies>

<dependency>

<groupId>org.springframework.boot</groupId>

<artifactId>**spring-boot-starter-web**</artifactId>

</dependency>

<dependency>

<groupId>org.springframework.boot</groupId>

<artifactId>**spring-boot-starter-thymeleaf**</artifactId>

</dependency>

<dependency>

<groupId>org.webjars</groupId>

<artifactId>**bootstrap**</artifactId>

<version>4.3.1</version>

</dependency>



```
<dependency>
<groupId>org.webjars</groupId>
<artifactId>jquery</artifactId>
<version>3.4.1</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>

<build>
<finalName>myjlcapp</finalName>
<plugins>

<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>
</project>
```



Spring MVC with Boot – WAR Model

- ♦ You can Develop as WAR file with Spring Boot
- ♦ WAR file is deployable application which can be deployed in any External Container to run.

How to deploy War file to Tomcat8

- 1) Make Sure that JDK8 is Installed.

```
E:\JDK1.8.0
E:\JDK1.8.0\bin
E:\JDK1.8.0\lib
```

- 2) Set JAVA_HOME

```
set JAVA_HOME=E:\JDK1.8.0
```

- 3) Make Sure that Tomcat8.5 is Installed.

```
E:\Tomcat8.5
E:\Tomcat8.5\bin\tomcat8
E:\Tomcat8.5\lib\*.jar
E:\Tomcat8.5\conf\server.xml
E:\Tomcat8.5\conf\web.xml
E:\Tomcat8.5\conf\tomcat-users.xml *
```

- 4) Update server.xml file

```
<Server port="21" shutdown="SHUTDOWN">

<Connector port="5050" protocol="HTTP/1.1"
    connectionTimeout="20000"
    redirectPort="5443" />
```

- 5) Make sure that Maven is Installed.

A) download the Maven from <https://maven.apache.org/download.cgi>
apache-maven-3.8.2-bin.zip

B) Extract the Zip and Copy to E:\ drive.

```
E:\apache-maven-3.8.2
E:\apache-maven-3.8.2\bin\mvn
E:\apache-maven-3.8.2\lib\*.jar
E:\apache-maven-3.8.2\conf\settings.xml
```



6) Set Maven Path

```
set path=%path%; E:\apache-maven-3.8.2\bin;
```

7) Add an user with roles manager-gui and manager-script in **tomcat-users.xml**

```
<user username="srinivas"  
      password="srinivas"  
      roles="manager-gui,manager-script" />
```

8) Add above Tomcat's user in the Maven setting file,
(**Maven will use this user to login Tomcat server**).

```
<server>  
<id>jlcTomcatServer</id>  
<username>srinivas</username>  
<password>srinivas</password>  
</server>
```

9) Add Tomcat7 Maven Plugin in pom.xml

```
<plugin>  
<groupId>org.apache.tomcat.maven</groupId>  
<artifactId>tomcat7-maven-plugin</artifactId>  
<version>2.2</version>  
<configuration>  
<url>http://localhost:5050/manager/text</url>  
<server>jlcTomcatServer</server>  
<path>/myjlcapp</path>  
</configuration>  
</plugin>
```

demo3-mvc-boot: Spring MVC with Boot (WAR)

Working Steps:

- 1) Copy **demo1-mvc-boot** as **demo3-mvc-boot**.
- 2) Update application.properties

No Properties Required

- 3) Update **MyBootApplication.java**



4) Update pom.xml

A) Add following 6 Dependencies

spring-boot-starter-web
jstl
standard
bootstrap
jquery
spring-boot-devtools

B) Add following 2 Plug-Ins

maven-compiler-plugin
tomcat7-maven-plugin

5) Open Command Line and Start the Tomcat8.5

E:\Tomcat8.5\bin>tomcat8

6) Open Another Command Line and run the Maven Command

...\demo3-mvc-boot>**mvn clean compile package tomcat7:deploy**

7) Open Browser and type <http://localhost:5050/myjlcapp>

Note : Maven Useful Commands

Deploy command => mvn tomcat7:deploy

UnDeploy command => mvn tomcat7:undeploy

ReDeploy command => mvn tomcat7:redploy

demo3-mvc-boot: Files Required

1. index.jsp	Same as Demo1
2. booksList.jsp	Same as Demo1
3. Book.java	Same as Demo1
4. IndexController.java	Same as Demo1
5. BooksController.java	Same as Demo1
6. JLCWebConfig.java	Same as Demo1
7. MyBootApplication.java	Updated in Demo3
8. application.properties	Updated in Demo3
9. pom.xml	Updated in Demo3



Welcome to CourseCube !!!

Latest Books from Srinivas Dande

Book Id	Book Name	Author	Price	Publications		
101	Master Spring Boot	Srinivas Dande	10000.0	JLC	View Book	Edit Book
102	Master MicroServices	Srinivas Dande	10000.0	JLC	View Book	Edit Book
103	Master Angular	Srinivas Dande	10000.0	JLC	View Book	Edit Book
104	Master React	Srinivas Dande	10000.0	JLC	View Book	Edit Book
105	Master Java Full Stack	Srinivas Dande	10000.0	JLC	View Book	Edit Book

7)MyBootApplication.java

```
package com.jlcindia.springboot;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.boot.builder.SpringApplicationBuilder;
import org.springframework.boot.web.servlet.support.SpringBootServletInitializer;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@SpringBootApplication
public class MyBootApplication extends SpringBootServletInitializer{

    public static void main(String[] args) {
        SpringApplication.run(MyBootApplication.class, args);
    }
    @Override
    protected SpringApplicationBuilder configure(SpringApplicationBuilder builder) {
        return builder.sources(MyBootApplication.class);
    }
}
```

8)application.properties

```
#No Properties Required
```



9)pom.xml

```
<project ...>
<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.springboot</groupId>
<artifactId>demo3-mvc-boot</artifactId>

<version>1.0</version>
<packaging>war</packaging>
<name>demo3-mvc-boot</name>
<description>This is My Spring MVC App with Boot</description>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.5.3</version>
</parent>

<dependencies>
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>

<dependency>
<groupId>jstl</groupId>
<artifactId>jstl</artifactId>
<version>1.2</version>
</dependency>

<dependency>
<groupId>taglibs</groupId>
<artifactId>standard</artifactId>
<version>1.1.2</version>
</dependency>

<dependency>
<groupId>org.webjars</groupId>
<artifactId>bootstrap</artifactId>
<version>4.3.1</version>
</dependency>
```



```
<dependency>
<groupId>org.webjars</groupId>
<artifactId>jquery</artifactId>
<version>3.4.1</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>
<build>
<finalName>myjlcapp</finalName>
<plugins>
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>
</plugin>

<plugin>
<groupId>org.apache.tomcat.maven</groupId>
<artifactId>tomcat7-maven-plugin</artifactId>
<version>2.2</version>
<configuration>
<url>http://localhost:5050/manager/text</url>
<server>jlcTomcatServer</server>
<path>/myjlcapp</path>
<update>true</update>
</configuration>
</plugin>
</plugins>

</build>
</project>
```



Spring Boot-MVC + JPA (BookStore- CURD Operationss)

demo4-mvc-data-jpa-boot: Files required

1. index.jsp	2. booksList.jsp
3. viewBook.jsp	4. addEditBook.jsp
5. IndexController.java	6. BookController.java
7. BookService.java	8. BookServiceImpl.java
9. BookDAO.java	10. Book.java
11. JLCWebConfig.java	12. MyBootApplication.java
13. application.properties	14. pom.xml

index.jsp

Welcome to JLC Bookstore
Best Books and Best Videos

[Display All Books](#)

bookList.jsp

Welcome to JLC Bookstore
Best Books and Best Videos

Book ID	Book Name	Author	Price	Category	Publications	Add New Book	
103	Master Angular7	Vas	800	Web	JLC123	Edit	Delete
106	Learn Spring5	Srinivas	500	Java	JLC	Edit	Delete
107	Learn Hibernate5	Srinivas	400	Java	JLC	Edit	Delete
108	Master Angular	Vas	300	Web	JLC	Edit	Delete



addEditBook.jsp

Welcome to JLC Bookstore
Best Books and Best Videos

Add New Book

Book Name

Author

Price

Category

Publications

Add New Book

addEditBook.jsp

Welcome to JLC Bookstore
Best Books and Best Videos

Update Book

Book Name

Master Angular7

Author

Vas

Price

800

Category

Web

Publications

JLC123

Update Book



viewBook.jsp

Welcome to JLC Bookstore

Best Books and Best Videos

Book Details	
Book Id	103
Book Name	Master Angular7
Author	Vas
Price	800
Category	Web
Publications	JLC123

[Goto Book Home](#)

1. index.jsp

```
<html>
<head>
<title>JLC Bookstore</title>
<link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<div class="card-header">
<h2 class="text-center">Welcome to JLC Bookstore</h2>
<h4 class="text-center"> Best Books and Best Videos</h4>
</div>
<br/><br/>
<div class="container" align="center">
<h2> <a href="showAllBooks">Display All Books</a> </h2>
</div>
</body>
</html>
```



2. bookList.jsp

```
<%@ taglib uri="http://www.springframework.org/tags/form" prefix="form" %>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>

<html>
<head>
<title>JLC Bookstore</title>
<link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<div class="card-header">
<h2 class="text-center">Welcome to JLC Bookstore</h2>
<h4 class="text-center"> Best Books and Best Videos</h4>
</div>
<br/><br/>
<div class="container" >
<table class="table table-striped table-bordered" style="font-size:20px;font-weight:bold;">
<tr>
<th>Book ID</th>
<th>Book Name</th>
<th>Author</th>
<th>Price</th>
<th>Category</th>
<th>Publications </th>
<th colspan="2" align="center">
<form:form action="addEditBookForm">
<input type="hidden" name="bookId" value="0"/>
<input type="submit" value=" Add New Book " class="btn btn-success btn-lg"/>
</form:form> </th>
</tr>
<c:forEach var="mybook" items="${MyBooksList}">
<tr>
<td> <a href="viewBook?bookId=${mybook.bid }"> ${mybook.bid } </a> </td>
<td>${mybook.bname }</td>
<td>${mybook.author }</td>
<td>${mybook.price }</td>
<td>${mybook.category }</td>
<td>${mybook.pub }</td>
<td>
<form:form action="addEditBookForm">
<input type="hidden" name="bookId" value="${mybook.bid }"/>
<input type="submit" value=" Edit " class="btn btn-primary btn-lg"/>
</form:form>
</td>
</tr>
</c:forEach>
</div>
```



```
<td>
<form:form action="deleteBook">
<input type="hidden" name="bookId" value="{mybook.bid }"/>
<input type="submit" value=" Delete " class="btn btn-danger btn-lg"/>
</form:form>
</td>
</tr>
</c:forEach>
</table>
</div>
</body>
</html>
```

3. viewBook.jsp

```
<%@ taglib prefix="form" uri="http://www.springframework.org/tags/form"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>

<html> <head>
<title>JLC Bookstore</title>
<link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<div class="card-header">
<h2 class="text-center">Welcome to JLC Bookstore</h2>
<h4 class="text-center">Best Books and Best Videos</h4>
</div>
<br/><br/>
<div class="container" >
<table class="table table-striped table-bordered" style="font-size:20px;font-weight:bold;">
<tr>
<td align="center" colspan="3"> Book Details</td>
</tr>
<tr>
<td> Book Id</td>
<td> ${MyBook.bid} </td>
</tr>
<tr>
<td> Book Name</td>
<td> ${MyBook.bname} </td>
</tr>
<tr>
<td> Author</td>
<td> ${MyBook.author} </td>
</tr>
```



```
<tr>
<td>Price</td>
<td>${MyBook.price} </td>
</tr>
<tr>
<td>Category</td>
<td>${MyBook.category} </td>
</tr>
<tr>
<td>Publications</td>
<td>${MyBook.pub} </td>
</tr>
</table> </div>
<div align="center">
<h2> <a href="showAllBooks">Goto Book Home</a> </h2>
</div>
</body> </html>
```

4. addEditBook.jsp

```
<%@ taglib prefix="form" uri="http://www.springframework.org/tags/form"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>

<html>
<head>
<title>JLC Bookstore</title>
<link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<div class="card-header">
<h2 class="text-center">Welcome to JLC Bookstore</h2>
<h4 class="text-center"> Best Books and Best Videos</h4>
</div>
<br/><br/>
<div align="center" class="container">
<form:form action="saveUpdateBook" method="post" modelAttribute="mybook">
<table class="table" style="font-size:20px;font-weight:bold;">
<tr>

<c:if test="${OpType eq 'ADD' }">
<td align="center" colspan="3"> <h2> Add New Book </h2></td>
</c:if>
<c:if test="${OpType eq 'UPDATE' }">
<td align="center" colspan="3"> <h2> Update Book </h2></td>
</c:if>
```



```
</tr>
<tr>
<td> Book Name</td>
<td><form:input path="bname" class="form-control form-control-lg"/></td>
<td><font color=red size=4><form:errors path="bname" /></font></td>
</tr>
<tr>
<td>Author</td>
<td><form:input path="author" class="form-control form-control-lg"/></td>
<td><font color=red size=4><form:errors path="author" /></font></td>
</tr>
<tr>
<td>Price</td>
<td><form:input path="price" class="form-control form-control-lg"/></td>
<td><font color=red size=4><form:errors path="price" /></font></td>
</tr>
<tr>
<td>Category</td>
<td><form:input path="category" class="form-control form-control-lg"/></td>
<td><font color=red size=4><form:errors path="category" /></font></td>
</tr>
<tr>
<td>Publications</td>
<td><form:input path="pub" class="form-control form-control-lg"/></td>
<td><font color=red size=4><form:errors path="pub" /></font></td>
</tr>
<tr>
<td align="center" colspan="3">
<input type="hidden" name="OpType" value="{OpType}"/>
<c:if test="{OpType eq 'ADD' }">
<input type="submit" value="Add New Book" class="btn btn-primary btn-lg"/>
</c:if>
<c:if test="{OpType eq 'UPDATE' }">
<input type="submit" value="Update Book" class="btn btn-primary btn-lg"/>
<form:hidden path="bid" />
</c:if>
</td>
</tr>
</table>
</form:form>
</div>
</body>
</html>
```



5. IndexController.java

```
package com.jlcindia.springboot.controller;

import org.springframework.stereotype.Controller;
import org.springframework.web.bind.annotation.GetMapping;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@Controller
public class IndexController {

    @GetMapping("/")
    public String showIndexPage() {
        System.out.println("-----showIndexPage-----");
        return "index";
    }

}
```

6. BookController.java

```
package com.jlcindia.springboot.controller;

import java.util.List;
import javax.servlet.http.HttpServletRequest;
import javax.validation.Valid;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller;
import org.springframework.ui.Model;
import org.springframework.validation.BindingResult;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.ModelAttribute;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestParam;
import org.springframework.web.bind.annotation.SessionAttributes;
import com.jlcindia.springboot.entity.Book;
import com.jlcindia.springboot.service.BookService;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@Controller
@SessionAttributes("MyBooksList")
public class BookController {

    @Autowired
    BookService bookService;
```



```
@GetMapping("/showAllBooks")
public String showBooksList(Model model) {
    System.out.println("-----BookController--showBooksList()-----");
    List<Book> blist=bookService.getAllBooks();
    model.addAttribute("MyBooksList", blist);

    return "booksList";
}

@PostMapping("/addEditBookForm")
public String addBookForm(@RequestParam("bookId") Integer bookId,Model model) {
    System.out.println("-----BookController--addEditBookForm()-----");
    System.out.println(bookId);
    Book book=new Book();
    String opType="ADD";
    if(bookId!=0) {
        book=bookService.getBookById(bookId);
        opType="UPDATE";
    }
    model.addAttribute("mybook", book);
    model.addAttribute("OpType", opType);

    return "addEditBook";
}

@PostMapping("/saveUpdateBook")
public String saveUpdateBook(@Valid @ModelAttribute("mybook") Book
book,BindingResult result,Model model,HttpServletRequest req) {
    System.out.println("-----BookController--saveUpdateBook()-----");
    String opType=req.getParameter("OpType");
    System.out.println(opType);
    if(opType.equals("ADD")) {
        bookService.addBook(book);
    }
    if(opType.equals("UPDATE")) {
        bookService.updateBook(book);
    }

    List<Book> blist=bookService.getAllBooks();
    model.addAttribute("MyBooksList", blist);

    return "booksList";
}
```



```
@PostMapping("/deleteBook")
public String deleteBook(@RequestParam("bookId") Integer bookId,Model model) {
    System.out.println("-----BookController--deleteBook()-----");
    bookService.deleteBook(bookId);
    List<Book> blist=bookService.getAllBooks();
    model.addAttribute("MyBooksList", blist);

    return "booksList";
}

@GetMapping("/viewBook")
public String viewBook(@RequestParam("bookId") String bookId,Model model) {
    System.out.println("-----BookController--viewBook()-----");
    System.out.println(bookId);
    Book book=bookService.getBookById(Integer.parseInt(bookId));
    model.addAttribute("MyBook", book);

    return "viewBook";
}
}
```

7. BookService.java

```
package com.jlcindia.springboot.service;

import java.util.List;
import com.jlcindia.springboot.entity.Book;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
public interface BookService {
    public List<Book> getAllBooks();
    public Book getBookById(Integer bid);
    public void addBook(Book book);
    public void updateBook(Book book);
    public void deleteBook(Integer bid);
}
```

8. BookServiceImpl.java

```
package com.jlcindia.springboot.service;

import java.util.List;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;
import com.jlcindia.springboot.dao.BookDAO;
import com.jlcindia.springboot.entity.Book;
```



```
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 **/

@Service
public class BookServiceImpl implements BookService{

    @Autowired
    BookDAO bookDAO;

    @Override
    public List<Book> getAllBooks() {
        System.out.println("-----BookServiceImpl--getAllBooks()-----");
        return bookDAO.findAll();
    }

    @Override
    public Book getBookById(Integer bid) {
        System.out.println("-----BookServiceImpl--getBookById()-----");
        return bookDAO.findById(bid).get();
    }

    @Override
    public void addBook(Book book) {
        System.out.println("-----BookServiceImpl--addBook()-----");
        bookDAO.save(book);
    }

    @Override
    public void updateBook(Book book) {
        System.out.println("-----BookServiceImpl--updateBook()-----");
        bookDAO.saveAndFlush(book);
    }

    @Override
    public void deleteBook(Integer bid) {
        System.out.println("-----BookServiceImpl--deleteBook()-----");
        bookDAO.deleteById(bid);
    }

}
```



9. BookDAO.java

```
package com.jlcindia.springboot.dao;

import org.springframework.data.jpa.repository.JpaRepository;
import com.jlcindia.springboot.entity.Book;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

public interface BookDAO extends JpaRepository<Book, Integer>{

}
```

10. Book.java

```
package com.jlcindia.springboot.entity;

import java.math.BigDecimal;
import javax.persistence.*;
import javax.validation.constraints.Max;
import javax.validation.constraints.Min;
import javax.validation.constraints.NotEmpty;
import javax.validation.constraints.NotNull;
import javax.validation.constraints.Size;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@Entity
@Table(name="mybooks")
public class Book {
    @Id
    @Column(name="bid")
    @GeneratedValue(strategy = GenerationType.AUTO)
    private Integer bid;

    @Column(name="bname")
    @NotEmpty(message = "Book Name is Required")
    @Size(min=5,max=20,message="Name length must be between 5 and 20")
    private String bname;

    @Column(name="author")
    @NotEmpty(message = "Author is Required")
    @Size(min=3,max=10,message="Name length must be between 3 and 10")
    private String author;

    @Column(name="price")
    @NotNull(message = "Price is Required")
```



```
@Min(value=500, message="Price must be min : 500")
@Max(value=1500, message="Price must be max : 1500")
private BigDecimal price;

@Column(name="category")
@NotEmpty(message = "Category is Required")
private String category;

@Column(name="pub")
@NotEmpty(message = "Pub is Required")
private String pub;

public Book() {
}
//Constrcutors
//Setters and Getters

@Override
public String toString() {
return "[" + bid + " , " + bname + " , " + author + " , " + price + " , " + category + " , " + pub+ " ]";
}
}
```

11. JLCWebConfig.java

```
package com.jlcindia.springboot;

import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.annotation.Bean;
import org.springframework.web.servlet.config.annotation.ResourceHandlerRegistry;
import org.springframework.web.servlet.config.annotation.WebMvcConfigurer;
import org.springframework.web.servlet.view.InternalResourceViewResolver;
import org.springframework.web.servlet.view.JstlView;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@SpringBootApplication
public class JLCWebConfig implements WebMvcConfigurer{

@Bean
public InternalResourceViewResolver viewResolver() {
InternalResourceViewResolver viewResolver = new InternalResourceViewResolver();
viewResolver.setViewClass(JstlView.class);
viewResolver.setPrefix("/WEB-INF/myjsps/");
viewResolver.setSuffix(".jsp");
return viewResolver;
}
}
```



```
@Override
public void addResourceHandlers(ResourceHandlerRegistry registry) {
    registry.addResourceHandler("/webjars/**").addResourceLocations("classpath:/META-INF/resources/webjars/");
}
}
```

12. MyBootApplication.java

```
package com.jlcindia.springboot;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@SpringBootApplication
public class MyBootApplication {
    public static void main(String[] args) {
        SpringApplication.run(MyBootApplication.class, args);
    }
}
```

13. application.properties

```
server.port=5599

logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n

spring.datasource.driver-class-name=com.mysql.jdbc.Driver
spring.datasource.url=jdbc:mysql://localhost:3306/bookstoredb
spring.datasource.username=root
spring.datasource.password=srinivas

spring.datasource.hikari.connectionTimeout=20000
spring.datasource.hikari.maximumPoolSize=5

spring.jpa.show-sql=true
spring.jpa.hibernate.ddl-auto=update
```



14. pom.xml

```
<project ....>
<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.springboot</groupId>
<artifactId>demo4-mvc-data-jpa-boot</artifactId>
<version>1.0</version>

<packaging>jar</packaging>
<name>demo4-mvc-data-jpa-boot</name>
<description>Spring MVC - Data JPA - BOOT - CURD Operations </description>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.5.3</version>
</parent>

<dependencies>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>

<dependency>
<groupId>org.apache.tomcat.embed</groupId>
<artifactId>tomcat-embed-jasper</artifactId>
</dependency>

<dependency>
<groupId>org.webjars</groupId>
<artifactId>bootstrap</artifactId>
<version>4.3.1</version>
</dependency>

<dependency>
<groupId>org.webjars</groupId>
<artifactId>jquery</artifactId>
<version>3.4.1</version>
</dependency>

<dependency>
<groupId>jstl</groupId>
<artifactId>jstl</artifactId>
<version>1.2</version>
</dependency>
```



```
<dependency>
<groupId>taglibs</groupId>
<artifactId>standard</artifactId>
<version>1.1.2</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-validation</artifactId>
</dependency>

<dependency>
<groupId>mysql</groupId>
<artifactId>mysql-connector-java</artifactId>
<version>5.1.32</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-mail</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>

<build>
<finalName>myjlcapp</finalName>
<plugins>

<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>
</project>
```

Spring Boot support following three embedded containers:-

- 1) Tomcat (default)
- 2) Jetty
- 3) Undertow.

1)Using Tomcat

- Tomcat is the Default Servlet Container in Spring Boot.
- Following spring-boot-starter-web brings Tomcat as Default.

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-web</artifactId>
</dependency>
```

2)Using Jetty

- Exclude Tomcat
- Include Jetty

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-web</artifactId>
  <exclusions>
    <exclusion>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-tomcat</artifactId>
    </exclusion>
  </exclusions>
</dependency>
```

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-jetty</artifactId>
</dependency>
```



3)Using Undertow

- Exclude Tomcat
- Include Undertow

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-web</artifactId>
  <exclusions>
    <exclusion>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-tomcat</artifactId>
    </exclusion>
  </exclusions>
</dependency>

<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-undertow</artifactId>
</dependency>
```

demo5-mvc-boot: Spring MVC with Boot (using Jetty)

Working Steps:

- 1) Copy **demo1-mvc-boot** as **demo5-mvc-boot**
- 2) Remove the WEB-INF under src/main/webapp (So that all the JSP's will be deleted)
- 3) Create **META-INF** folder under src/main/resources
- 4) Create **resources** folder under src/main/resources/META-INF
- 5) Create **WEB-INF** under src/main/resources/META-INF/resouces
- 6) Create **myhtmls** folder under src/main/resources/META-INF/resources/WEB-INF
- 7) Place index.html and booksList.html under **myhtmls**
 - a)index.html (new in Demo5)
 - b)booksList.html (new in Demo5)
- 8) Update **ViewResolver** Bean from JLCWebConfig.java

```
resolver.setPrefix("/WEB-INF/myhtmls/");
resolver.setSuffix(".html");
```



9) Update **addResourceHandlers()** method from JLCWebConfig.java

```
registry.addResourceHandler("/webjars/**")
    .addResourceLocations("classpath:/META-INF/resources/webjars/");

registry.addResourceHandler("/WEB-INF/**")
    .addResourceLocations("classpath:/META-INF/resources/WEB-INF/");
```

10) Update **BookController.java**

```
@Controller
public class BookController {

    @GetMapping("/showBooks")
    public String getBooks() {
        System.out.println(" BooksController - getBooks()");

        return "booksList";
    }
}
```

11) Remove **Bookjava**

12)Add the following property in application.properties

```
server.port=12345
```

13)Remove the following dependencies from pom.xml

```
<dependency>
    <groupId>org.apache.tomcat.embed</groupId>
    <artifactId>tomcat-embed-jasper</artifactId>
</dependency>
```

14)Exclude **Tomcat**

```
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
    <exclusions>
        <exclusion>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-tomcat</artifactId>
        </exclusion>
    </exclusions>
</dependency>
```



15) Add the dependency for **Jetty** in pom.xml

```
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-jetty</artifactId>
</dependency>
```

16) Make sure that all the Files as given below are updated correctly.

17) Run the Boot Application

18) Open the browser and hit <http://localhost:12345/>

demo5-mvc-boot: Files Required

1. index.html	Updated in Demo5
2. booksList.html	Updated in Demo5
3. IndexController.java	Same as Demo1
4. BooksController.java	Updated in Demo5
5. JLCWebConfig.java	Updated in Demo5
6. MyBootApplication.java	Same as Demo1
7. application.properties	Updated in Demo5
8. pom.xml	Updated in Demo5

1) index.html

```
<!DOCTYPE html>
<html>
<head>
<title> JLCBookStore</title>
<link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<h1 class="text-center"> Welcome to Java Learning Center !!!</h1>
<br/>
<h2 class="text-center"> <a href="showBooks"> Show Books </a></h2>

<script src="webjars/jquery/3.4.1/jquery.min.js"></script>
<script src="webjars/bootstrap/4.3.1/js/bootstrap.min.js"></script>

</body>
</html>
```



2) BooksList.html

```
<!DOCTYPE html>
<html>
<head>
<title> JLCBookStore</title>
<link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<div class="container">
<b1 class="text-center"> Welcome to Java Learning Center !!!</b1>

<h2 class="text-center"> Books List Here </h2>
<h2 class="text-center"> Books List Here </h2>
<h2 class="text-center"> Books List Here </h2>
<h2 class="text-center"> Books List Here </h2>
<h2 class="text-center"> Books List Here </h2>
<h2 class="text-center"> Books List Here </h2>

</div>
<script src="webjars/jquery/3.4.1/jquery.min.js"></script>
<script src="webjars/bootstrap/4.3.1/js/bootstrap.min.js"></script>

</body>
</html>
```

4) BookController.java

```
package com.jlcindia.springboot;

import org.springframework.stereotype.Controller;
import org.springframework.web.bind.annotation.GetMapping;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@Controller
public class BookController {

    @GetMapping("/showBooks")
    public String getBooks() {
        System.out.println(" BooksController - getBooks()");

        return "booksList";
    }
}
```



5) JLCWebConfig.java

```
package com.jlcindia.springboot;

import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.annotation.Bean;
import org.springframework.web.servlet.ViewResolver;
import org.springframework.web.servlet.config.annotation.ResourceHandlerRegistry;
import org.springframework.web.servlet.config.annotation.WebMvcConfigurer;
import org.springframework.web.servlet.view.InternalResourceViewResolver;
import org.springframework.web.servlet.view.JstlView;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@SpringBootApplication
public class JLCWebConfig implements WebMvcConfigurer{

    @Bean
    public ViewResolver viewResolver() {
        InternalResourceViewResolver resolver = new InternalResourceViewResolver();
        resolver.setPrefix("/WEB-INF/myhtmls/");
        resolver.setSuffix(".html");
        resolver.setViewClass(JstlView.class);
        return resolver;
    }

    @Override
    public void addResourceHandlers(ResourceHandlerRegistry registry) {
        registry.addResourceHandler("/webjars/**")
        .addResourceLocations("classpath:/META-INF/resources/webjars/");

        registry.addResourceHandler("/WEB-INF/**")
        .addResourceLocations("classpath:/META-INF/resources/WEB-INF/");
    }
}
```

7)application.properties

```
server.port=12345
```



8)pom.xml

```
<project ...>
<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.springboot</groupId>
<artifactId>demo5-mvc-boot</artifactId>
<version>1.0</version>
<name>demo5-mvc-boot</name>
<description>This is My Spring MVC App with Boot</description>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.5.3</version>
</parent>

<dependencies>
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
<exclusions>
<exclusion>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-tomcat</artifactId>
</exclusion>
</exclusions>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-jetty</artifactId>
</dependency>

<dependency>
<groupId>jstl</groupId>
<artifactId>jstl</artifactId>
<version>1.2</version>
</dependency>

<dependency>
<groupId>>taglibs</groupId>
<artifactId>standard</artifactId>
<version>1.1.2</version>
</dependency>
```



```
<dependency>
<groupId>org.webjars</groupId>
<artifactId>bootstrap</artifactId>
<version>4.3.1</version>
</dependency>

<dependency>
<groupId>org.webjars</groupId>
<artifactId>jquery</artifactId>
<version>3.4.1</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>

<build>
<finalName>myjlcapp</finalName>
<plugins>

<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>
</project>
```